



CS 520: Introduction to Quantum Computing

Course Code	CS 520
Course Name	Introduction to Quantum Computing
L-T-P-C	2-1-0-3
Prerequisites	Basic Linear Algebra and Probability
Intended for	M.Tech/Ph.D./B.Tech 3rd/4th year.
Mutual Exclusion	Other Research Methodology Courses
Approval	57th BoA (Oct 2024); 47th Senate (Sep 2025), in the Minor in Quantum Technology curriculum

Course Contents

- **Introduction** : Review of Linear Algebra, Probability, Qubits, Interference, Superposition, Entanglement, Young double slit, Mach-Zehnder Experiment, Dirac Notation, Postulates of Quantum Mechanics. [8 hours]
- **Quantum Computation** : Universal Quantum gates, CNOT gate, Tofoli gate, Fredkin gate, No-cloning theorem, Basic Deutsch's algorithm, Quantum Teleportation, BB84, Bell States, Quantum Circuits [8 hours]
- **Quantum Algorithms** : Quantum parallelism, Deutsch's algorithm, Deutsch-Jozsa algorithm, Quantum Search algorithms, Grover's search, Shor's algorithm, Speeding up the solutions of NP-complete problems, Quantum Supremacy [8 hours]
- **Quantum Cryptography** : Quantum key distribution, Simon's algorithm, and Cryptanalysis.[4 hours]

Text Books

1. M A Nielsen and I L Chuang, **Quantum Computation and Quantum Information**.
2. P Ka , R Laflamme and M Mosca, **An Introduction to Quantum Computing**.

References

1. https://www.cse.iitk.ac.in/users/rmittal/course_s24.php
2. <http://books.google.ca/books?id=qYHTvHPvmG8C>